

Activity 6 — Rapid Prototyping with AI — From Concept to Testable Artefact in One Hour

Course	Fast-Track Innovations with Agile Design Thinking and Generative AI (GenAI)
WSQ Course Code	TGS-2024049781
Topic	Topic 02: Problem Framing and Ideation: Leveraging Design Thinking and Generative AI
Duration	50 minutes · Group size: 3–5 participants
Tools	Design Thinking Studio, ChatGPT or Microsoft Copilot, Adobe Firefly (optional)

Learning Outcome Addressed

LO2 · LO4 · A4 — Prototyping with AI-driven tools: rapid development and feedback.

The Real Case

A hospital innovation team in Singapore needed to test whether a self-service kiosk would reduce queue times at a specialist outpatient clinic. The traditional route — a scoped software build — would have taken a quarter and a budget approval. Instead the team used a GenAI tool to draft the screen copy and flow, printed the screens on paper, and ran a 'Wizard of Oz' test in the clinic corridor: a staff member sat behind a curtain and swapped the paper screens by hand while real patients used it. Within two days they learned that elderly patients did not fail at the touchscreen — they abandoned at the point where the kiosk asked for an NRIC, because they were not sure whether it was safe to enter it in public view. That single insight redirected the entire design, and it cost nothing but paper and two afternoons. Had they built the software first, they would have discovered it after the budget was spent.

Your Scenario

Take the top-voted concept from Activity 5. You have 50 minutes to make it testable by a real person — not to build it.

What You Will Do

Learners use GenAI to draft prototype content and user flows, assemble a low-fidelity prototype, and define an explicit test plan with falsifiable assumptions — practising 'build to think' rather than 'build to ship'.

What you will produce

A low-fidelity prototype (paper, slide or wireframe) with a written test plan, listed assumptions and the criteria

that would prove the concept wrong.

Step-by-Step Instructions

1. Open the Prototype stage in the Design Thinking Studio workspace.
2. Post your concept in one paragraph into Prototype Description.
3. Prompt the AI: 'Draft the user flow and screen-by-screen copy for [your concept]. Flag the three steps most likely to cause drop-off.'
4. Post the flow into User Flow and the key screens into Screens / Wireframes.
5. List every assumption your concept depends on in Assumptions — be ruthless.
6. Optionally use Adobe Firefly to generate a visual for the concept.
7. Move to the Test stage and write a Test Plan: who you would test with, the task you would set, and what result would prove you wrong.
8. Swap workspaces with another group, run their prototype as a user, and post honest feedback into their User Feedback section.

Discussion Questions

1. The kiosk team learned the real barrier was privacy, not usability. Which of your assumptions, if wrong, would sink your concept fastest?
2. What is the cheapest possible artefact that would still generate a real reaction from a real user?
3. 'Build to think, not to ship.' What is the risk of a prototype that looks too polished?
4. Where did GenAI genuinely save time here, and where would relying on it have hidden the privacy insight?
5. Write the one question you would ask a test user that could prove your concept wrong.

How You Know You Are Done

Your prototype can be run by someone from another group without you explaining it, and your test plan states in advance what result would falsify the concept.

Debrief — What the Room Should Conclude

A prototype's purpose is to buy information at the lowest possible price. The kiosk team bought a quarter's worth of learning for two afternoons and a stack of paper. Note the specific trap in polished prototypes: when an artefact looks finished, test users critique surface details and politely withhold fundamental objections — a rough paper screen invites honest reaction in a way a clean mockup does not. GenAI is genuinely excellent at the drafting layer here: screen copy, flow logic, edge cases, even the interview script. What it cannot do is stand in the corridor and watch a 74-year-old hesitate over her NRIC. Every prototype should ship with its falsification criteria written down in advance; a test you cannot fail is a demonstration, not a test.